

Two-Tower Recommender

Technical report on two-stage recommendation retrieval, ranking, and offline evidence review.

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1. Study Abstract

This project documents a recommendation stack where retrieval and ranking are deliberately separated so candidate generation can scale without giving up downstream business relevance. The combination of two-tower retrieval, ANN candidate generation, and a ranker gives the project a realistic online shape. The measured uplift over popularity and retrieval-only ranking makes the business value concrete. The implementation evidence is drawn from committed repository artefacts, benchmark outputs, automated tests, and generated screenshots.

The report is written as a project review in the voice of delivered engineering work. Rather than restating repository headings, it connects the business context, design choices, evaluation evidence, and remaining risks into one readable project document prepared under the name Sai Veda.

Keywords: Retrieval + ranking, trained from scratch, implementation evidence, benchmark results, project validation

2. Recommendation Context

Recommendation quality breaks when systems either over-index on popularity or overfit a ranking model that cannot scale to a full catalog. The architecture therefore mirrors production: first recall, then precision. A complete, offline, deterministic recommender system: a from-scratch two-tower retrieval model, ANN candidate generation, a gradient-boosted ranker, and a leak-free temporal evaluation harness - trained on millions of implicit interactions and architected for 1B. No GPU, no network, no paid APIs.

2.1 Retrieval-Then-Ranking Problem Frame

The central project problem can be stated directly. Recommendation quality breaks when systems either over-index on popularity or overfit a ranking model that cannot scale to a full catalog. The architecture therefore mirrors production: first recall, then precision. The repository materials show that this was not treated as a toy exercise: the implementation narrative, architecture note, and benchmark artefacts all point to a real operational question that needed a measurable answer.

2.2 Assessment Questions

The document review was guided by a small set of concrete questions. Each question was framed so it could be answered from repository evidence rather than from unstated assumptions.

Research question: Does the implemented project address the stated technical problem in a complete and reviewable manner?

Hypothesis: The repository design, architecture notes, and implementation artefacts are expected to demonstrate end-to-end coverage of the core project objective.

Research question: Do the benchmark and test artefacts support the main performance or quality claim?

Hypothesis: The recorded evidence is expected to support the headline result reported for this project: beats popularity +76% recall; ranker +7.8% nDCG.

Research question: Can the results be reviewed and reproduced from committed repository evidence?

Hypothesis: The presence of 33 automated tests, benchmark outputs, and generated screenshots is expected to provide a transparent validation path.

3. Recommendation Foundations

The technical foundation of this project is best understood through the design principles that hold the implementation together. Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance.

3.1 Recommendation Foundations

- 1 Retrieval and ranking are separated intentionally because they solve different scale problems. Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance.
- 2 Temporal evaluation is non-negotiable for recommendation systems that serve future behavior.
- 3 The item embedding space is treated as infrastructure that downstream ranking can build on.

3.2 Repository Evidence Notes

This document covers the design decisions, the trade-offs behind each stage, and the concrete path from the measured single-box run to a 1B-interaction / 100M-item production system. Retrieval and ranking are split because they optimise different things: Scoring the full catalogue with a heavy model per request is infeasible at 100M items; scoring only the ANN shortlist is.

- Temporal data split and training statistics.
- Two-tower retrieval model and item embedding space.
- ANN candidate recall plus gradient-boosted reranking.

4. Serving Architecture And Training Flow

Candidate recall and ranking precision are intentionally split so the serving path scales without giving up business relevance. Retrieval and ranking are split because they optimise different things: Scoring the full catalogue with a heavy model per request is infeasible at 100M items; scoring only the ANN shortlist is.

4.1 Modeling Procedure

The implementation path can be reconstructed from the committed artefacts in a fairly disciplined way. Interaction modeling: Started with a temporal split so evaluation reflects forward-looking recommendation behavior. Embedding training: Trained user and item towers with in-batch negatives and popularity correction. Two-stage serving path: Separated ANN recall from a richer ranking stage that can use additional features. Outcome review: Measured recall lift over popularity and ranking lift over retrieval-only baselines.

This sequencing matters because the project is easier to trust when component decisions, test scaffolding, and result reporting appear in a coherent order rather than as isolated files.

5. Evaluation Setup And Artefacts

5.1 Repository Evidence Base

Source: committed repository artefacts including implementation code, automated tests, benchmark outputs, architecture notes, and generated visual evidence. Coverage: 33 automated tests, 0 benchmark table(s), and 4 screenshot asset(s) were available for document preparation. Schema / artefact types: README overview, ARCHITECTURE design note, benchmark markdown and CSV outputs, test suites, and figure assets generated from the project run path. Availability: all evidence cited in this document is sourced from the repository itself.

5.2 Evaluation Protocol

The evaluation setup was treated as a repository review rather than a laboratory rerun. Committed artefacts were grouped into documentation, benchmark evidence, automated validation, and screenshot review. In practical terms this meant examining 0 benchmark table(s), 4 screenshot asset(s), and 33 automated tests while reading the implementation narrative in sequence. The review lens stayed aligned with the project goal: Train user and item towers from scratch on implicit interactions. Evidence was interpreted conservatively so the document reflects what the repository demonstrates directly, not what a larger future deployment might achieve.

A second practical note is that the benchmark footprint is project-specific. The benchmark evidence reviewed for this report includes `benchmarks/benchmark_scale.py`, `benchmarks/metrics.csv`, `benchmarks/summary.json`.

6. Performance Findings

6.1 Benchmark Summary

The principal repository claim for this project is recorded as: beats popularity +76% recall; ranker +7.8% nDCG. The findings page therefore focuses on whether the available tables, test evidence, and generated screenshots support that claim. The combination of two-tower retrieval, ANN candidate generation, and a ranker gives the project a realistic online shape. The measured uplift over popularity and retrieval-only ranking makes the business value concrete. The retrieval gains matter because they increase the candidate pool quality before ranking begins. The ranker lift is easier to explain when shown as an improvement over a strong retrieval baseline.

6.2 Interpretation

The retrieval gains matter because they increase the candidate pool quality before ranking begins. The ranker lift is easier to explain when shown as an improvement over a strong retrieval baseline. The dashboard evidence makes clear that recall and nDCG improve for different reasons and should be reviewed separately.

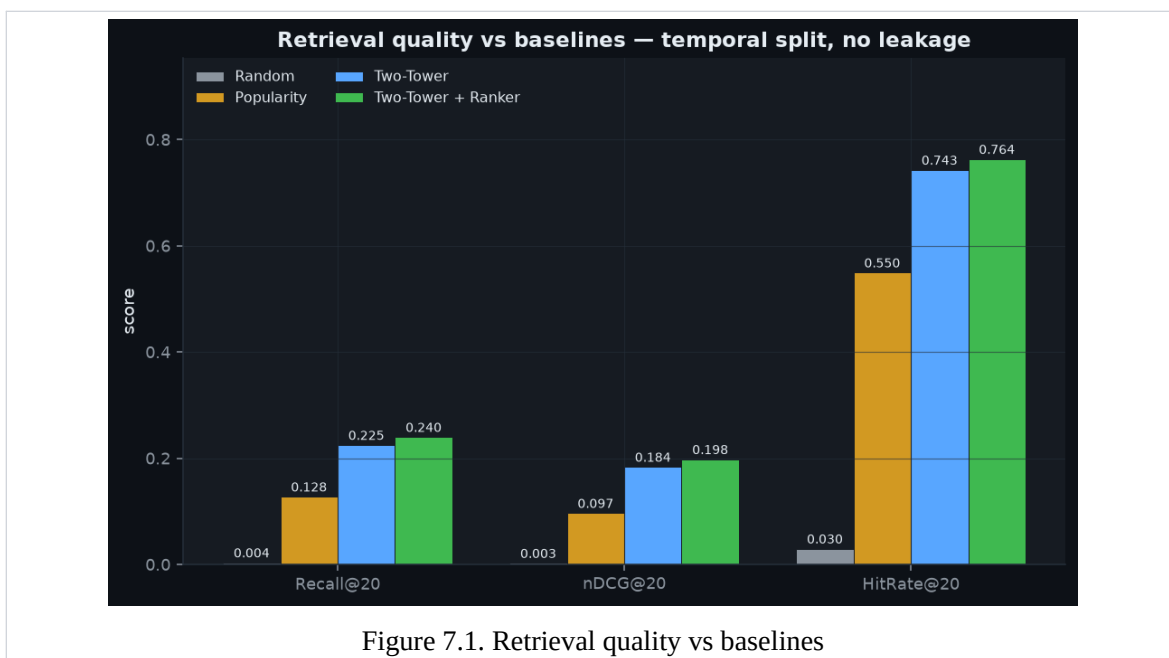
The benchmark table is not treated as a marketing surface. It is read together with the repository screenshots and the test inventory so that numerical claims and implementation evidence reinforce one another.

7. Recommendation Figures

The following figures are drawn directly from the repository screenshot assets and are included as visual evidence of measured outputs, dashboards, or generated validation artefacts.

7.1 Retrieval quality vs baselines

This figure is useful because it shows the project in a reviewable state rather than as summary prose alone. The retrieval gains matter because they increase the candidate pool quality before ranking begins.



7.2 Item embedding space (PCA), coloured by category

The visual evidence attached to Item embedding space (PCA), coloured by category helps connect the quantitative story to the implemented workflow. The ranker lift is easier to explain when shown as an improvement over a strong retrieval baseline.

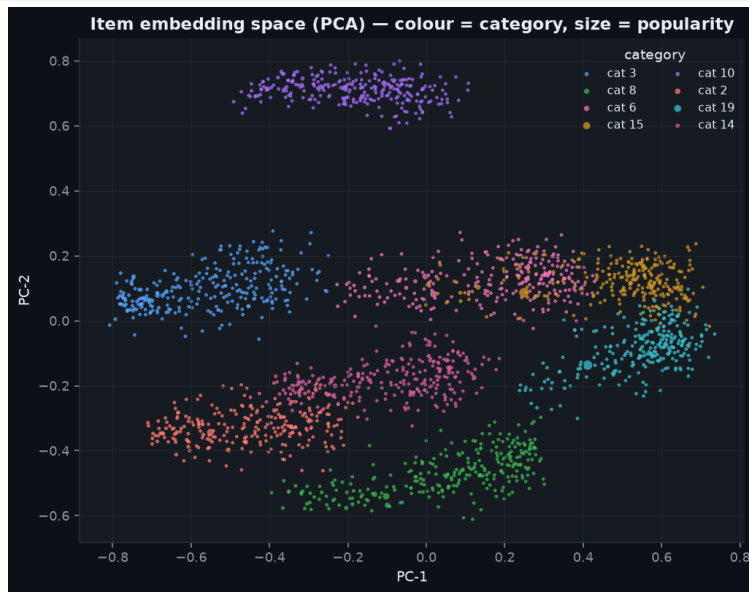


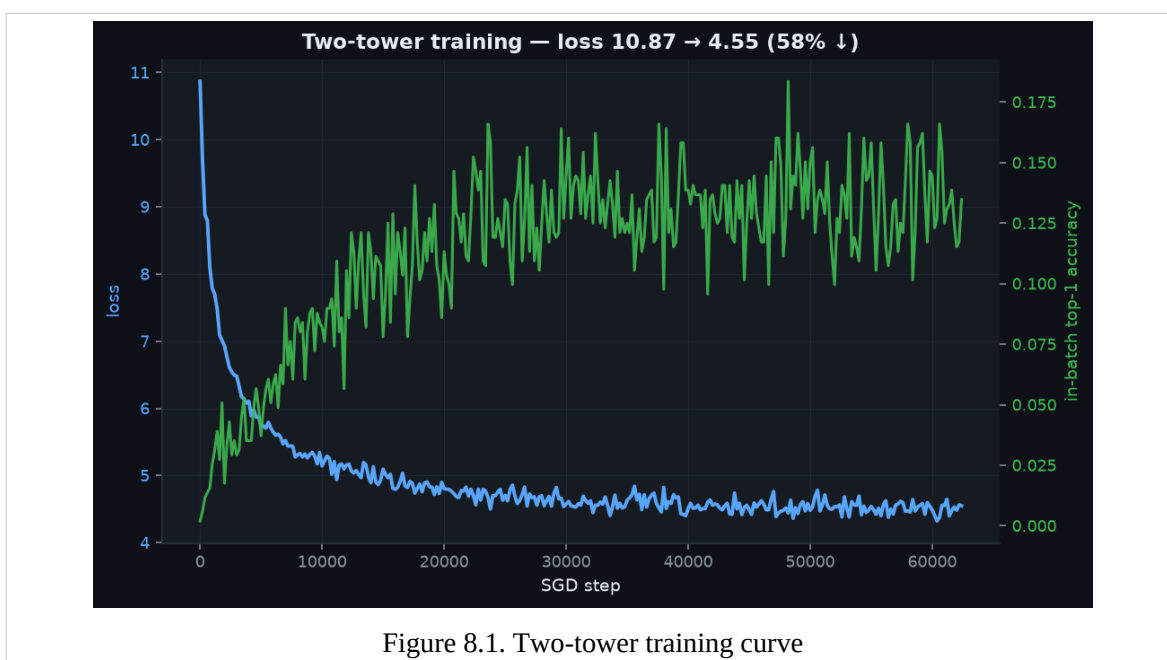
Figure 7.2. Item embedding space (PCA), coloured by category

8. Validation Coverage

The second figure set focuses on validation continuity. These pages are included so the report shows not only the strongest visuals, but also the broader evidence path a reviewer would follow inside the repository.

8.1 Two-tower training curve

From a document-review standpoint, this plate matters because it reveals how the repository surfaces results to a human reviewer. The dashboard evidence makes clear that recall and nDCG improve for different reasons and should be reviewed separately.



8.2 Offline evaluation dashboard

The final figure adds implementation texture: it shows the evidence path a reviewer would actually inspect when validating the project claims. Addressed by using ANN for candidate recall instead of full-catalog scoring.

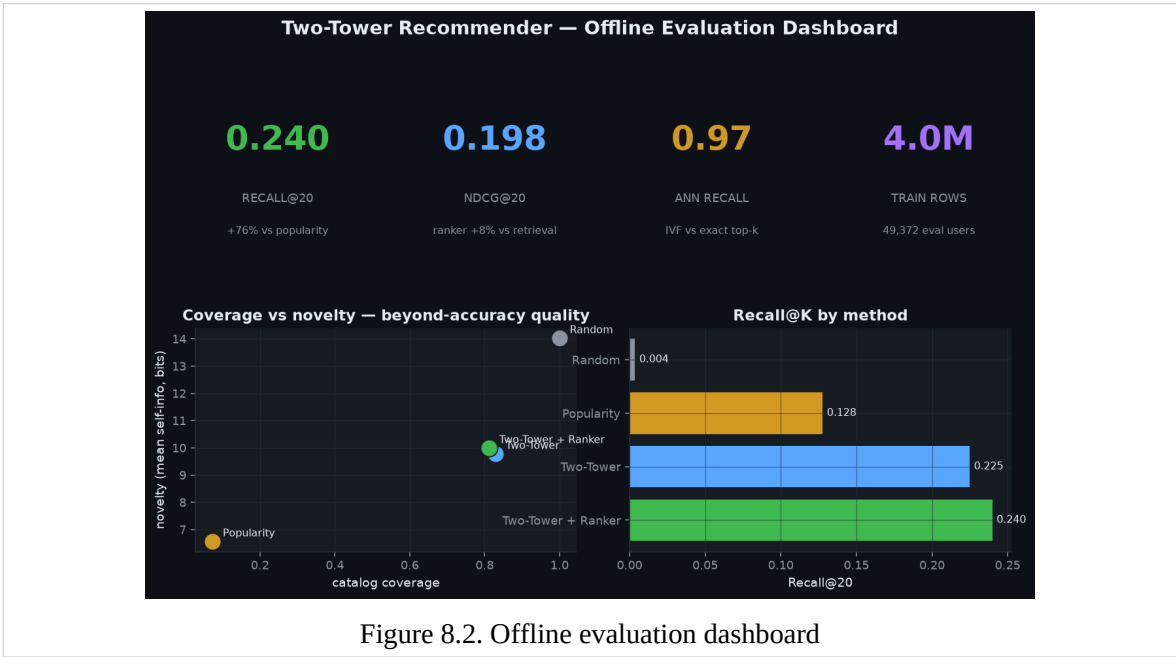


Figure 8.2. Offline evaluation dashboard

8.3 Validation Checklist

Item	Status	Evidence
Automated tests	Available	33 tests committed in repository validation flow.
Benchmark results	Limited	No benchmark markdown tables were present; discussion relies on screenshots and h
Screenshots	Available	4 screenshot asset(s) included in the repository evidence set.
Architecture note	Available	ARCHITECTURE.md documents design choices, scale assumptions, and trade-offs.

9. System Discussion

The combination of two-tower retrieval, ANN candidate generation, and a ranker gives the project a realistic online shape. The measured uplift over popularity and retrieval-only ranking makes the business value concrete. The retrieval gains matter because they increase the candidate pool quality before ranking begins. The ranker lift is easier to explain when shown as an improvement over a strong retrieval baseline. The dashboard evidence makes clear that recall and nDCG improve for different reasons and should be reviewed separately. The analysis indicates that the repository is strongest where implementation, benchmark artefacts, and screenshots point to the same conclusion.

9.1 Present Limits

Catalog scale: Addressed by using ANN for candidate recall instead of full-catalog scoring.
Evaluation leakage: Mitigated through a strict temporal split and feature discipline.
Over-reliance on popularity: Reduced by combining learned embeddings with richer ranking features.

These limitations do not invalidate the project. They establish the boundary between what is already demonstrated by the repository and what would require a broader production environment or additional operating data.

10. Assessment Close

This project document concludes that Two-Tower Recommender is best understood as a complete technical implementation supported by repository-native evidence. This project documents a recommendation stack where retrieval and ranking are deliberately separated so candidate generation can scale without giving up downstream business relevance.

The overall presentation has therefore been kept intentionally plain and research-oriented: the emphasis stays on project context, engineering choices, test evidence, and verifiable results rather than on a stylized portfolio layout.

10.1 Further Work

- Add session-aware or context-aware retrieval features for short-term intent shifts.
- Experiment with harder negative sampling and refreshed item embeddings.
- Introduce business-constraint reranking for diversity, margin, or inventory goals.

11. Appendix Material

11.1 Repository Evidence Inventory

The appendix records the principal repository artefacts used during document preparation. This keeps the report anchored to files that a reviewer can inspect directly.

Artefact	Category	Purpose
README.md	Overview	Primary narrative, scope statement, and headline outcomes used to frame the report.
ARCHITECTURE.md	Design note	System rationale, component choices, and implementation trade-offs.
benchmarks/benchmark_scale.py	Benchmark	Quantitative result or execution artefact used in the findings section.
benchmarks/metrics.csv	Benchmark	Quantitative result or execution artefact used in the findings section.
benchmarks/summary.json	Benchmark	Quantitative result or execution artefact used in the findings section.
assets/metrics_vs_baselines.png	Screenshot	Visual validation surface referenced as 'Retrieval quality vs baselines' in the document review.
assets/embedding_space.png	Screenshot	Visual validation surface referenced as 'Item embedding space (PCA), coloured by category' in the document review.
assets/training_loss.png	Screenshot	Visual validation surface referenced as 'Two-tower training curve' in the document review.
assets/kpi_dashboard.png	Screenshot	Visual validation surface referenced as 'Offline evaluation dashboard' in the document review.
tests/	Validation	Automated test suite containing 33 committed tests used to support implementation claims.

11.2 Internal References

1. README.md - primary repository overview and headline results.
2. ARCHITECTURE.md - system design, implementation rationale, and scale notes.
3. benchmarks/benchmark_scale.py - benchmark or result artefact used in the analysis section.
4. benchmarks/metrics.csv - benchmark or result artefact used in the analysis section.
5. benchmarks/summary.json - benchmark or result artefact used in the analysis section.

6. tests/ - automated validation suite used to support implementation claims.
7. assets/embedding_space.png - generated screenshot evidence included in the report.
8. assets/kpi_dashboard.png - generated screenshot evidence included in the report.